

DISCRETE MATHEMATICS

LECTURE 12

0.1. **Hamming codes.** These are perfect code with separation 3. Hence the control matrix of a Hamming code contains multiples of *all* possible syndromes. This implies that any word is at distance at most 1 from some (unique) code word.

Example 0.1. Let us work over the field $\mathbb{Z}_2$, and let H consist of all words of length 3 except 000. That is,

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

(or some permutation of those columns).

An aside: the code is a linear $[7, 4]$ code. One can check that

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

We can correct the words $\mathbf{x}_1 = (1010111)$, $\mathbf{x}_2 = (0001010)$ and $\mathbf{x}_3 = (1011000)$ provided we know that they contain only one error. Using H we compute their syndromes

$$\mathbf{x}_1 H^T = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 H^T = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \text{and} \quad \mathbf{x}_3 H^T = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

We find them in columns 3, 1 and 5 of H , so corrections should be made by adjusting the symbols 3, 1, and 5 respectively. Thus the corrected words are (1000111), (1001010) and (1011100).

In general, when working over $\mathbb{Z}_2$, we can work with words of length n and a code of dimension m if and only if

$$\begin{aligned} n &= \text{number of columns of } H \\ &= \text{number of possible syndromes} \neq 0 \\ &= 2^{\text{length of a syndrome}} - 1 = 2^{n-m} - 1. \end{aligned}$$

(Length of a syndrome = number of rows in $H = n - m$.)

Setting $r = n - m$, this condition becomes

$$\begin{cases} n = 2^r - 1 \\ r = n - m \end{cases} \iff \begin{cases} n = 2^r - 1 \\ m = n - r = 2^r - 1 - r \end{cases}$$

This shows that we can construct binary Hamming codes of type $[2^r - 1, 2^r - 1 - r]$ for any $r \in \mathbb{N}$.

All Hamming codes are perfect because by making balls of radius one around every code word we include every word in the space. (Any word can be made into a code word by altering only one digit; that is the distance to the code word is only one. Also it can be seen by looking at H that the separation is 3.)

We can construct a Hamming code over any field:

Example 0.2. Here we work over the field $\mathbb{Z}_3$. The earlier example of a perfect code $G =$

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}$$

In $\mathbb{Z}_3^4$ it is a Hamming code with control matrix

$$H = \begin{pmatrix} 2 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}.$$

All words in $\mathbb{Z}_3^2$ apart from (00) (and hence all syndromes) are *multiples* of a column in H .

$$\begin{array}{cccccccc} 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ H_4 & 2H_4 & H_3 & 2H_1 & 2H_2 & 2H_3 & H_2 & H_1 \end{array}$$

If we receive $w = (1020)$ the syndrome is $\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 2H_2$. This means that the coset leader is $\mathbf{c} = (0200)$ and the corrected word is $\mathbf{w} - \mathbf{c} = (1120) \in C$.

Some other interesting examples of linear codes include Reed-Solomon codes (used in DVD-players for example) and Reed-Muller codes (includes for example, a $[32, 6]$ code of separation 16 used by Mariner 9 to send images of Mars back to Earth; the separation 16 makes it possible to correct as many as 7 errors).

0.1.1. *Vandermonde Matrices and Reed-Solomon codes.* Say we want to construct a code over a field F with a certain separation σ . If we build H such that any $\sigma - 1$ columns are linearly independent but some σ columns are dependent we will get separation σ by Theorem ??.

To construct H let H have $\sigma - 1$ rows and $n \geq \sigma$ columns where each column is of the form $(1 \beta_i \beta_i^2 \cdots \beta_i^{\sigma-2})^T$ for some $\beta_i \in F$. Choose different β_i for each column. Then

$$H = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \beta_1 & \beta_2 & \cdots & \beta_n \\ \beta_1^2 & \beta_2^2 & \cdots & \beta_n^2 \\ \vdots & \vdots & & \vdots \\ \beta_1^{\sigma-2} & \beta_2^{\sigma-2} & \cdots & \beta_n^{\sigma-2} \end{pmatrix} \quad \beta_i \neq \beta_j.$$

Clearly σ columns will be dependent since they are in a $(\sigma - 1)$ -dimensional vector space. We will show that any $\sigma - 1$ columns are independent. If they were dependent then the rows would also be dependent and this would mean we could find $c_0, c_1, \dots, c_{\sigma-2}$ not all zero such that

$$(0, 0, \dots, 0) = c_0(1, \dots, 1) + c_1(\beta_{i_1}, \dots, \beta_{i_{\sigma-1}}) + c_2(\beta_{i_1}^2, \dots, \beta_{i_{\sigma-1}}^2) + \cdots + c_{\sigma-2}(\beta_{i_1}^{\sigma-2}, \dots, \beta_{i_{\sigma-1}}^{\sigma-2})$$

Equivalently

$$c_0 + c_1\beta_{i_k} + c_2\beta_{i_k}^2 + \cdots + c_{\sigma-2}\beta_{i_k}^{\sigma-2} = 0$$

for all $\sigma - 1$ different k . Then we have $\sigma - 1$ roots to an equation of degree $\sigma - 2$, a contradiction. Our assumption of dependent columns must have been wrong.

This shows by Theorem 3.10 from Andersson that with the given control matrix H we have a code of separation σ .

Example 0.3. The smallest interesting example is when $\sigma = 3$. Assume for simplicity that $F = \mathbb{Z}_7$. We get codes of separation 3 using 2 rows and ≥ 3 columns of the type described to form H . For example,

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}.$$

This is a linear $[6, 4]$ code. We know we can correct one error since $\sigma = 3$. However it is quite easy to do so: If we want to send $(z_1, z_2, \dots, z_6)$ and the receiver gets $\mathbf{w} = (z_1, \dots, z_{k-1}, z_k + e, z_{k+1}, \dots, z_6)$ instead then $\mathbf{w}$ has syndrome $\mathbf{w}H^T = \begin{pmatrix} e \\ ke \end{pmatrix}$, where e is the error and k is the position of the error.

For example if $\mathbf{w} = (210504)$ is received then the syndrome

$$\begin{pmatrix} 2 + 1 + 0 + 5 + 0 + 4 \\ 2 + 2 \cdot 1 + 3 \cdot 0 + 4 \cdot 5 + 5 \cdot 0 + 6 \cdot 4 \end{pmatrix} = \begin{pmatrix} 12 \\ 48 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} e \\ ke \end{pmatrix}$$

if and only if

$$\begin{cases} e = 5 \\ k = 6e^{-1} = 6 \cdot 5^{-1} = 6 \cdot 3 = 18 = 4. \end{cases}$$

The coset leader has 5 in position 4: $\mathbf{c} = (000500)$ and the corrected word is $\mathbf{w} - \mathbf{c} = (210004)$.

If we want to correct 2 errors we need $\sigma = 5$ so we must add two rows to H :

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 1^2 & 2^2 & 3^2 & 4^2 & 5^2 & 6^2 \\ 1^3 & 2^3 & 3^3 & 4^3 & 5^3 & 6^3 \end{pmatrix}.$$

This gives a code with fewer words (7^2 instead of 7^4) but higher capacity to correct errors.

Codes like in the above example where we make a column for each non-zero element in the field are called Reed-Solomon codes (used in DVD-players for example).

0.1.2. *Recursive construction of codes and Reed-Muller codes.* Two linear codes can be combined to form a new code:

Theorem 0.4. If C_1 and C_2 are $[n, m_1]$ and $[n, m_2]$ codes with separation σ_1 and σ_2 , then

$$C = \{(\mathbf{x}, \mathbf{x} + \mathbf{y}) \in F^{2n} \mid \mathbf{x} \in C_1, \mathbf{y} \in C_2\}$$

is a linear $[2n, m_1 + m_2]$ code with separation $\sigma = \min\{2\sigma_1, \sigma_2\}$.

Proof. It is straightforward to check that C is a linear subspace and that given bases $\mathbf{e}_1, \dots, \mathbf{e}_{m_1}$ and $\mathbf{f}_1, \dots, \mathbf{f}_{m_2}$ of C_1 and C_2 respectively, the subspace C has a basis

$$\{(\mathbf{e}_1, \mathbf{e}_1), \dots, (\mathbf{e}_{m_1}, \mathbf{e}_{m_1}), (\mathbf{0}, \mathbf{f}_1), \dots, (\mathbf{0}, \mathbf{f}_{m_2})\}$$

and hence has dimension $m_1 + m_2$.

For the separation look at the weight of words in C . Among words of the form $(\mathbf{x}, \mathbf{x})$, where $\mathbf{x} \in C_1$, we have $w(\mathbf{x}, \mathbf{x}) = 2w(\mathbf{x})$ and the minimum value is $2\sigma_1$.

Among words $(\mathbf{x}, \mathbf{x} + \mathbf{y})$, with $\mathbf{y} \neq \mathbf{0}$, we have $w(\mathbf{x}, \mathbf{x} + \mathbf{y}) \geq w(\mathbf{y})$ because if $y_i \neq 0$ then either $x_i \neq 0$ or $x_i + y_i \neq 0$.

The minimum weight we get is σ_2 (by choosing $\mathbf{x} = \mathbf{0}$ and $\mathbf{y}$ of minimal weight). Hence $\sigma = \min\{2\sigma_1, \sigma_2\}$. $\square$

Let us use the above theorem to construct first order Reed-Muller codes.

Start with C_1 generated by (01, 10) in $\mathbb{Z}_2^2$ and C_2 generated by (11). Then $\sigma_1 = 1$ and $\sigma_2 = 2$.

Combining C_1 and C_2 as in Theorem 4.6 we obtain a code C_1^1 generated by (0101, 1010, 0011) with separation $\sigma_1^1 = \min\{2\sigma_1, \sigma_2\} = 1$.

Continue using $C_2^1 = \{0000, 1111\}$ and combine C_1^1 with C_2^1 . The new code C_1^4 is generated by

01010101
10101010
00110011 ,
00001111

which is a 4-dimensional code with separation $\sigma_1^4 = \min\{2\sigma_1^1, \sigma_2^1\} = 4$.

If we continue one more step we get a code C_1^{111} generated by

0101010101010101
1010101010101010
1111111111111111
0011001100110011
0000111100001111
0000000011111111

of separation 8 and finally a $[32, 6]$ code of separation 16 used by Mariner 9 to send images of Mars back to Earth. The separation 16 makes it possible to correct as many as 7 errors. The price we pay is sending 32 digits instead of 6 to describe elements in our 6-dimensional space of code words. That is $32 - 6 = 26$ extra digits.

0.1.3. Golay codes. In 1949 Golay constructed two perfect codes. One $[11, 6]$ code called $\mathcal{G}_{11}$ and one $[23, 12]$ code called $\mathcal{G}_{23}$.

One can show that there are no other perfect codes with more than two words and correcting more than one error!

Let us look at the smallest of the two Golay codes. First we investigate the code C . It has generating matrix:

$$G = \left[\begin{array}{c|cccc} & 011111 \\ & 101221 \\ & 110122 \\ I_6 & 121012 \\ & 122101 \\ & 112210 \end{array} \right];$$

the code is over $\mathbb{Z}_3$ so there are 3^6 words in the code.

Note that the last five rows of the right part of G are cyclic permutations of the sequence 01221.

One can check that all scalar products of rows are zero and hence the code is self-dual.

Because of self-duality and the fact that $\mathbb{Z}_3$ has the elements 0, 1 and $2 = -1$ shows that $0 \equiv \langle \mathbf{x}, \mathbf{x} \rangle \equiv \text{number of non-zero entries of } \mathbf{x} = w(\mathbf{w})$. Therefore every code word has a weight divisible by 3.

Once can check that no code word has weight 3 (details in the book p. 48). This implies that $d(C) \geq 6$. Obviously $d(C) \leq 6$ since for example the generating words have weight 6.

Thus we must have $d(C) = 6$. Removing the first column of A we now get an $[11, 6]$ code with separation 5. This is $\mathcal{G}_{11}$. We verify that it is perfect:

$$\frac{3^6}{\text{No. of code words}} \cdot \frac{\left(1 + \binom{11}{1} \cdot 2 + \binom{11}{2} \cdot 2^2\right)}{\text{size of ball of radius 2 around a word}} = \frac{3^{11}}{\text{total number of words in space}}.$$